KULDEEP GURJAR

Robotics & AI Engineer | ROS Expert

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🌐 KD | 🎓 Google Scholar | 💼 LinkedIn

SKILLS

- **ROBOTICS SOFTWARE**
  Robot Operating System (ROS1/ROS2), Nav2, MoveIt2, SLAM, Simulation(Gazebo, MuJuCo, Issac Sim)
- **PROGRAMMING & TOOLS**
  Python, C++, MatLab, PyTorch, Formal Methods, Git, server, Docker, TensorRT, ONNX, Fast API
- **ARTIFICIAL INTELLIGENCE**
  ML, DL, RL, LLM, VLM, STT
- **EMBEDDED COMPUTING BOARD**
  Nvidia Jetson Orin, RaspberryPi, Arduino
- **ROBOTS**
  Mobile robot(Quanser self-driving car, Turtlebot3, wheelchair), Manipulator (UR10, Upper body Robotic System-DRDO/Dual Arm), Biped (BRS1-DRDO)
- **MEDICAL & ROBOTIC SENSORS**
  ECG, EEG, EMG, RaDAR(FMCW), Camera(RealSense, OAK-D, RPi), LiDAR(RP, Livox)
- **RACING VEHICLES**
  Go-Kart, BAJA-ATV

PROFESSIONAL EXPERIENCE

ROS Software Engineer

IHub-Data, Hyderabad, India | February 2026 - Present

- Leading AI developments for autonomous wheelchair project.
- Involved in constrained environment navigation for autonomous wheelchair.
- Developed a glass detection framework and deployed on NVIDIA Jetson, leveraging Quant Turbo–based KV cache compression with a custom NVMap driver plugin for optimized memory management and real-time inference.
- Containerized the wheelchair navigation software stack and isolated the AI modules into separate Docker containers for modularity, resource efficiency, and reproducible build.
- Deployed a voice- and language-driven control interface for autonomous wheelchair navigation.

Data Scientist

Research & Development Establishment (Eng.), DRDO, Pune, India | May 2024 - January 2026

- Dual arm development and motion planning with custom planners, and MoveIt2.
- Implemented a Large Language Model-driven and object detection-based high-level action generator for dual-arm task planning.
- Developed admittance control framework for compliant dual-arm manipulation and force-torque feedback systems
- Stable dynamic walking of biped with Reinforcement learning.
- Involved in software stack development for upper-body and lower-body of humanoid including kinematics, dynamics, ROS2, and control.

Senior Research Fellow (SRF)

Defence Institute of Advanced Technology, Pune, India | July 2023 - May 2024

- Developed computer vision system for missile fall-of-shot estimation.

- Created novel radar-based vital signs classification system for non-invasive age and gender detection in children using machine learning
- Designed EEG-based brain-robot interface using ratio of band power (RBP) features for upper limb motor imagery classification
- Built automated drone state identification system using RF signal analysis with Random Forest classifier, demonstrating robustness across diverse environments

Research Intern

Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore, India | June 2022 - June 2023

- Developed reinforcement learning approach with Signal Temporal Logic (STL) specifications using funnel-based control for formal verification of tasks.
- Developed time-aware Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm for time-sensitive robotic tasks
- Implemented weighted-STL with Q-Learning for multi-objective task specification, enabling preference-based robot behavior

EDUCATION

Master of Technology in Automation and Robotics

Defence Institute of Advanced Technology (DIAT), Pune, India | August 2021 - May 2023

Bachelor of Engineering in Automobile Engineering

Madhav Institute of Technology and Science, Gwalior, India | June 2015 - May 2019

SELECTED PUBLICATIONS

- K. Gurjar, S. Agrawal, K. N. Faisal and R. R. Sharma, *"Classifying EEG-based Upper Limb Motor Imagery Tasks for Brain-Robot Interface-based System Development,"* 2024 International Conference on Advanced Robotics and Mechatronics (ICARM), Tokyo, Japan. **[Best Paper Nominee]**
- R. R. Sharma, K. Gurjar, et al., *"Advanced Wavelet Transform-Based Automated System for Drone State Identification Using RF Signal,"* JAIS, Vol. 21, No. 11 (2024), pp. 951-957.
- S. Kouloorkar, K. Gurjar, et al., *"ResNet50-based Automated Method for Monocular Depth Estimation using Video,"* 2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT).
- *"Non-invasive Age Classification of Children Using Radar-based Vital Sign Analysis"* [Under Review - Internet of Things and Cyber-Physical Systems]

AWARDS & HONORS

- **Best Paper Nominee - ICARM 2024 International Conference, Tokyo**
- MHRD Scholarship for M.Tech (2021-2023)
- State Government Scholarship for B.E. (2015-2019)
- Best Presentation Award - e-National Kart Racing Championship

TEACHING EXPERIENCE

Teaching Assistant - Programming Languages for Robotics

Defence Institute of Advanced Technology, Pune

Delivered hands-on tutorials on ROS2 fundamentals, robot kinematics/dynamics analysis, and Gazebo simulation for postgraduate students. Conducted practical sessions with TurtleBot3 hardware and developed course materials for autonomous navigation.